

MAJOR PROJECT PRESENTATION

NexFolio: An Explainable AI (XAI) Framework for Intelligent Portfolio Risk Profiling, Real-Time Market Analytics, and Institutional Tax Optimization

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TITLE JUSTIFICATION

“Explainable AI (XAI)”

Replaces opaque 'black-box' predictions with game-theoretic TreeSHAP feature attributions. Translates complex mathematical weights into human-understandable risk drivers.

“Portfolio Risk Profiling”

Multi-dimensional evaluation beyond simple P&L using 36 quantitative features across return momentum, annualized volatility, downside semi-variance, Sharpe/Sortino ratios, and market Beta.

“Real-Time Market Analytics”

Decouples sub-millisecond in-memory valuation from heavy analytics using a Dual-Loop architecture and a 5-state Market Data Pedigree state machine (Live, Delayed, Reference).

“Institutional Tax Optimization”

Implements native statutory compliance with the new Income-tax Act, 2025 (Tax Year 2026–27), modeling STCG @ 20%, LTCG @ 12.5%, buyback taxation, and an 8-year loss carryforward bank.

BACKGROUND

Quantitative Portfolio Management & AI

- Modern Portfolio Theory (MPT) & Multi-Factor Models
- Exponential growth of retail investors (160M+ Demat accounts)
- Shift from simple P&L tracking to risk-adjusted metrics
- Supervised ML classification for portfolio risk tiers
- Statutory transition to Income-tax Act, 2025 (TY 2026–27)

PORTFOLIO RISK & INTELLIGENCE ARCHITECTURE FLOW

End-to-End Quantitative Machine Learning & Explainability Pipeline

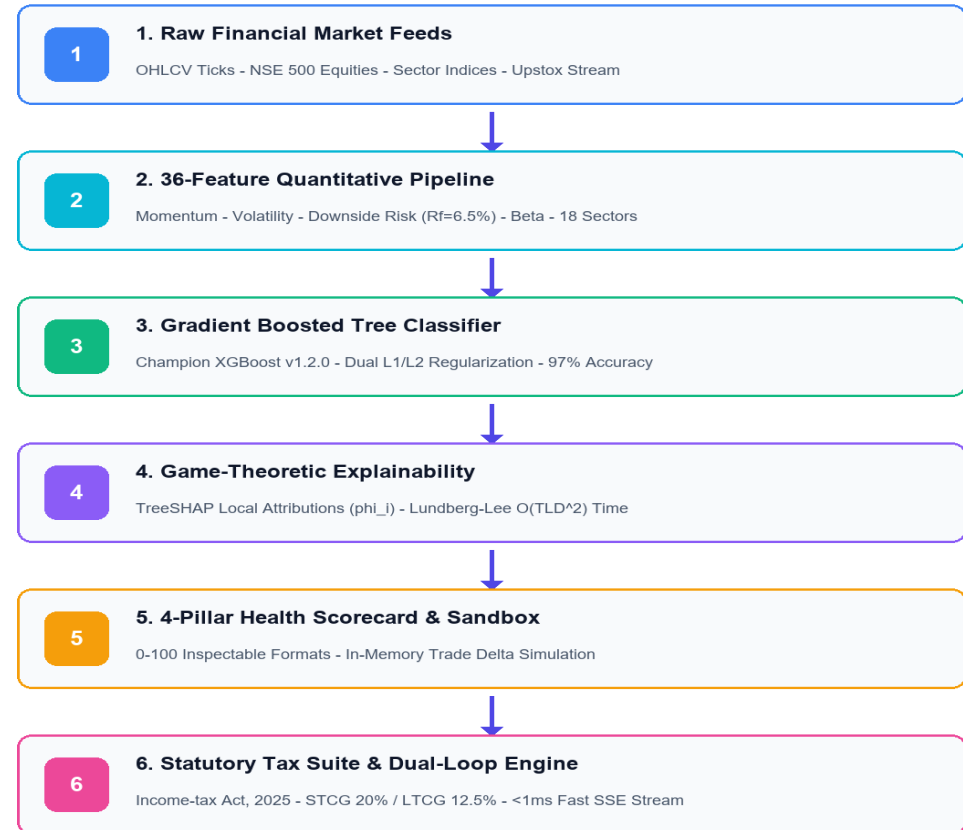


Fig 1: Quantitative Portfolio Intelligence Architecture

MOTIVATION

Retail Capital Exposure

Black-Box AI Distrust

Budget 2026-27 Tax Overhaul

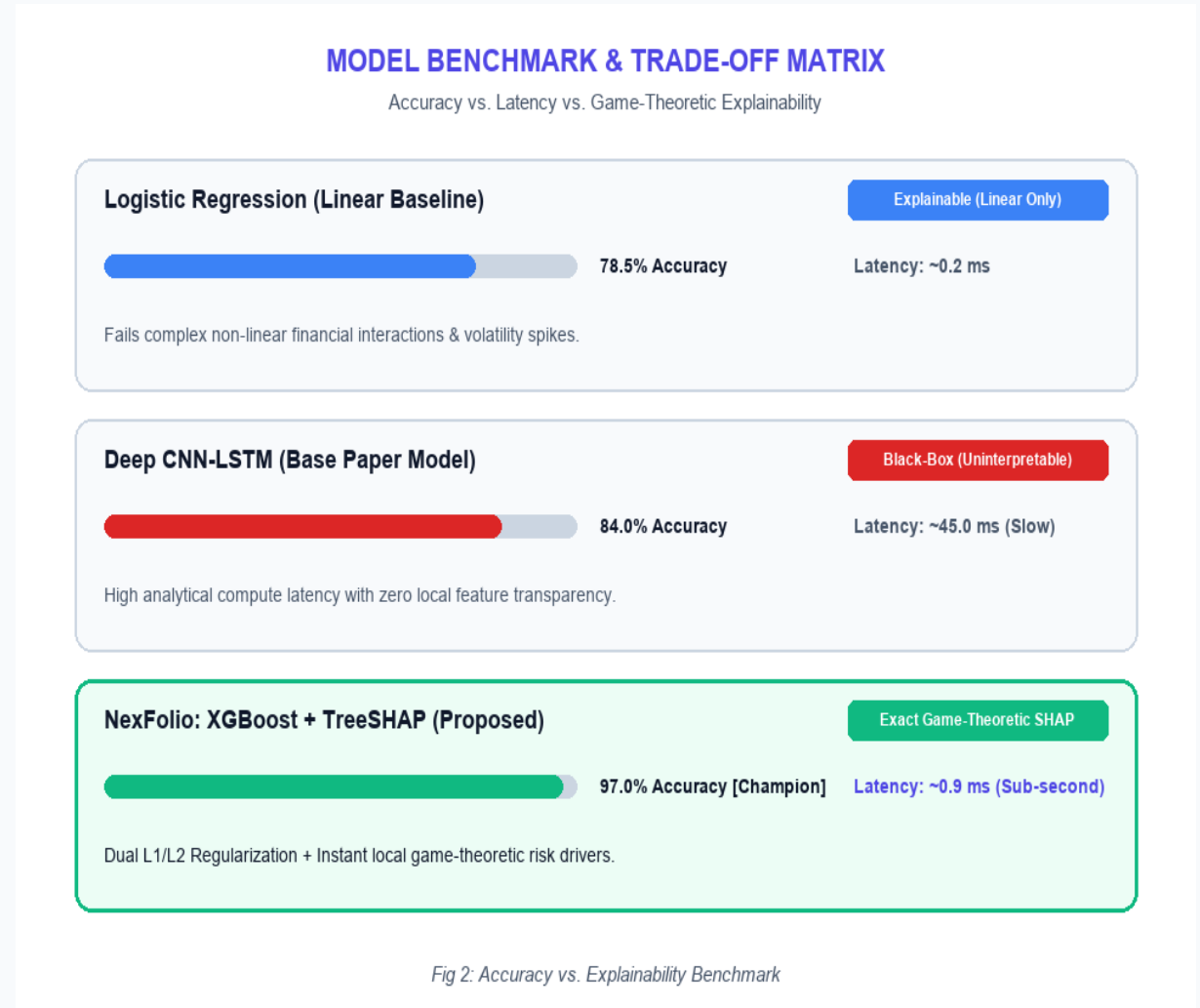
Cross-Sector Market Volatility

High-Latency System Freezes

Audit & Regulatory Filings

THE PORTFOLIO RISK & BLACK-BOX CHALLENGE

- Model Opacity in deep neural networks
- High feature multi-collinearity & noise
- Inference latency bottlenecks (>40ms)
- Outdated 365-day tax approximations



LITERATURE SURVEY

Authors	Title	Methods	Advantages	Disadvantages
P. Singh et al. (IEEE Access, 2023) [Base Paper]	Harnessing a Hybrid CNN-LSTM Model for Portfolio Performance	1D-CNN, LSTM, Markowitz Mean-Variance	Captures temporal trends; multi-scale feature learning	Opaque black-box; high compute latency (~45ms); no tax math.
I. Aruleba, Y. Sun (IEEE Access, 2024)	Effective Credit Risk Prediction Using Ensemble Classifiers	Random Forest, XGBoost, SMOTE-ENN, TreeSHAP	Ensemble superiority on tabular data; exact Shapley values	Binary credit default focus; lacks multi-class portfolio tiers.
I. Aruleba, Y. Sun (IEEE Access, 2025)	An Improved Ensemble Method With Data Resampling	Stacked Ensemble (RF, LR, CNN + MLP), SMOTE-ENN	Mitigates extreme class skew; reduces individual model variance	High stacking overhead; static batch testing without streaming.
Vijayanand, Smrithy (IDT Journal, 2025)	Explainable AI-Enhanced Ensemble Learning for Financial Fraud	Voting Ensemble (XGB, RF, DT) + TreeSHAP	Achieves 99.9% accuracy; provides feature-level audit trust	Limited to transaction fraud; lacks portfolio metrics or tax rules.

RESEARCH GAP



PROBLEM STATEMENT

THE CURRENT CHALLENGE:

- Model Opacity: Black-box deep neural networks obscure underlying risk drivers.
- Analytical Latency: Heavy compute bottlenecks (>40ms) stall live ticker streams.
- Statutory Ignorance: Outdated tools ignore tax drag under the new Income-tax Act, 2025.

OUR PROPOSED OBJECTIVE (NexFolio):

- Explainable ML: Regularized XGBoost (97% Acc) with game-theoretic TreeSHAP attributions.
- Dual-Loop Engine: Decouple <1ms live quote streaming from background deep analytics.
- Institutional Tax Suite: Native STCG (20%), LTCG (12.5%), and 8-year Loss Harvesting Bank.

OBJECTIVES

01

36-Feature Pipeline

Capture return momentum, volatility, downside risk, Beta, and 18 sector weights.

02

XGBoost & TreeSHAP

Train regularized gradient-boosted trees (>95% accuracy) with sub-second SHAP attributions.

03

4-Pillar Scorecard

Formulate an inspectable 0–100 score across Diversification, Volatility, Efficiency, and Drawdown.

04

What-If Sandbox

Enable in-memory trade simulations with instant delta calculations without database writes.

05

Dual-Loop Engine

Decouple <1ms live quote streaming from slow persistence with a 5-state Pedigree FSM.

06

Statutory Tax Suite

Implement Income-tax Act, 2025 STCG @ 20%, LTCG @ 12.5%, Buybacks, and 8-Year Loss Bank.

CONCLUSION

- Investigate black-box limitations in portfolio risk models and formulate an explainable 36-feature quantitative pipeline.
- Develop an institutional ML architecture combining regularized XGBoost with TreeSHAP for sub-second risk attributions.
- Formulate a deterministic 4-pillar health scorecard (0–100) and an in-memory What-If trade simulation sandbox.
- Implement a Dual-Loop valuation architecture to decouple <1ms live quote streaming from background analytics.
- Implement statutory tax compliance for the Income-tax Act, 2025 featuring STCG 20%, LTCG 12.5%, and an 8-year Loss Bank.

REFERENCES

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THANK YOU